

Tuesday Warm-up: Fields of Magnetized Objects, Auxiliary Field $\mathbf{H}$

Prof. Jordan C. Hanson

November 11, 2025

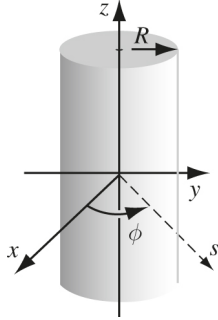


Figure 1: A cylinder with $\mathbf{M} = ks^2\hat{\mathbf{z}}$.

1 Memory Bank

- **Bound volume current density:**

$$\mathbf{J}_b = \nabla \times \mathbf{M} \quad (1)$$

- **Bound surface current density:**

$$\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}} \quad (2)$$

- **Auxiliary field, $\mathbf{H}$:**

$$\mathbf{H} = \frac{1}{\mu_0} \mathbf{B} - \mathbf{M} \quad (3)$$

- **Ampère's Law**, integral form:

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}} \quad (4)$$

- **Ampère's Law**, differential form:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} \quad (5)$$

2 Fields of Magnetized Objects

1. A very long circular cylinder of radius R carries a magnetization $\mathbf{M} = ks^2\hat{\phi}$, where k is a constant (Fig. 1). Find the magnetic field due to $\mathbf{M}$, for points inside and outside the cylinder. *Hint: what is the total current due to bound surface and volume current densities?*

2. In the previous exercise, what is the auxiliary field, $\mathbf{H}$?

3 The Auxiliary Field, $\mathbf{H}$

1. (a) Break the current density in Ampère's Law (differential form) into *bound* and *free* current densities. (b) Substitute the bound current density with its definition in terms of magnetization. (c) Gather the curls on one side of the formula to find Ampère's Law in matter. (d) Produce the integral form.
2. What is the *divergence* of $\mathbf{H}$?
3. An infinitely long cylinder of radius R carries a magnetization $\mathbf{M} = ks\hat{\mathbf{z}}$, where k is a constant, s is the distance from the axis, and $\mathbf{J}_f = 0$. (a) Locate the bound currents and calculate the fields. (b) Use Ampère's Law to find $\mathbf{H}$, and then obtain $\mathbf{B}$.